

Interactive Application to Support Learning Digital Systems to Embedded Systems

Computer Engineering and Informatics Project

M2 – Requirements, related work, viable solution

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Introduction

- Demand for interactive methodologies to enhance student engagement
- Lack of immediate feedback mechanisms outside the classroom
- Limitations of static and repetitive exercise formats
- Requirement for continuous knowledge validation and self-assessment



Related Work

Platform	Interactivity	Technical Domain	IA / Feedback
Duolingo	High	Languages	Adaptative
NotebookLM	High	Large (superficial)	Adaptative
Khan Academy	Medium	Maths & Science	Pre-defined tips
Our solution	High + Gamification	Digital to Embedded Systems (deep)	Adaptative



Users

Student

- Primary learner and content consumer;
- Focused on self-improvement and practice;
- Uses AI support to bridge knowledge gaps.

Professor

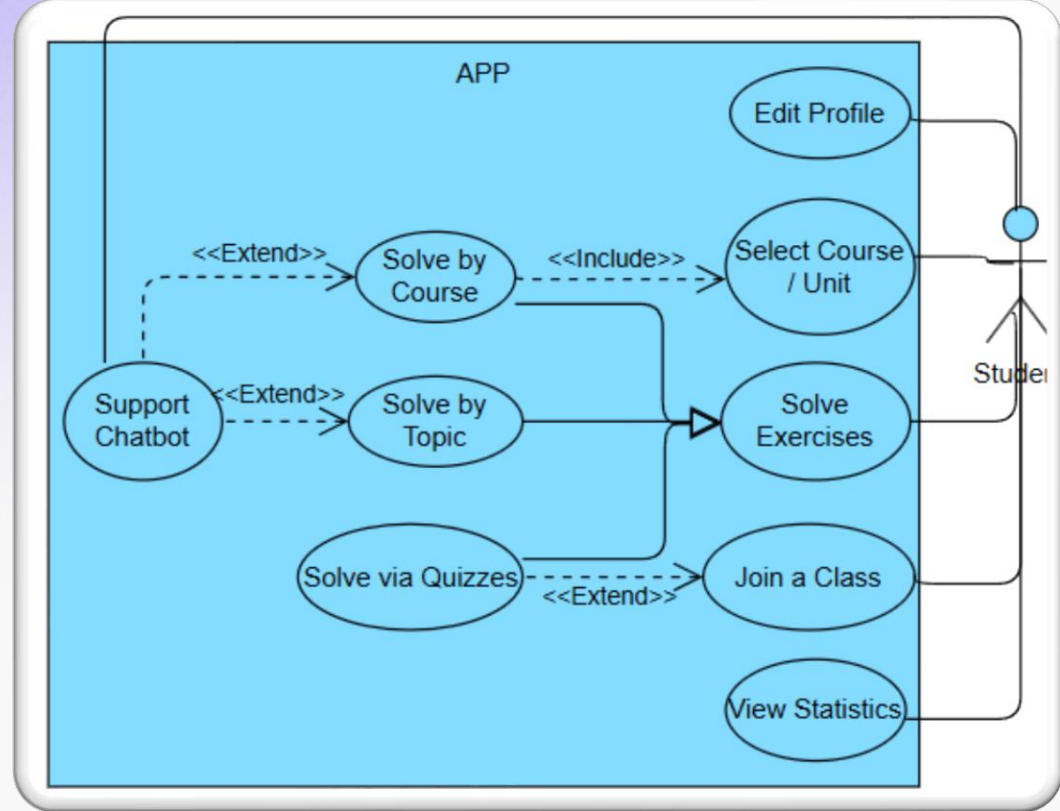
- Educational facilitator and mentor;
- Responsible for content curation and class guidance;
- Monitors student performance and engagement.

Admin

- System guardian and infrastructure manager;
- Controls access, permissions, and global settings;
- Ensures the platform's operational health.

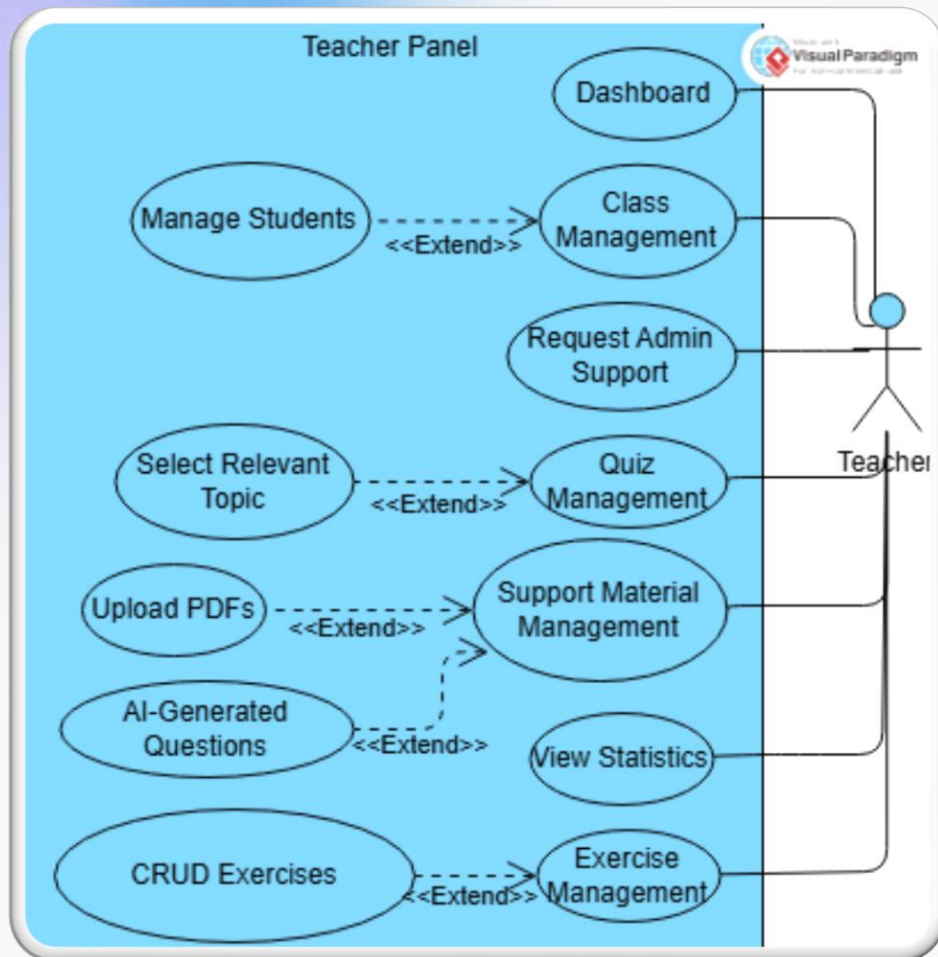
Student

- Accesses learning materials and exercises.
- Solves quizzes and practices by topic or course.
- Tracks personal progress and statistics.
- Interacts with the AI Chatbot for real-time support.



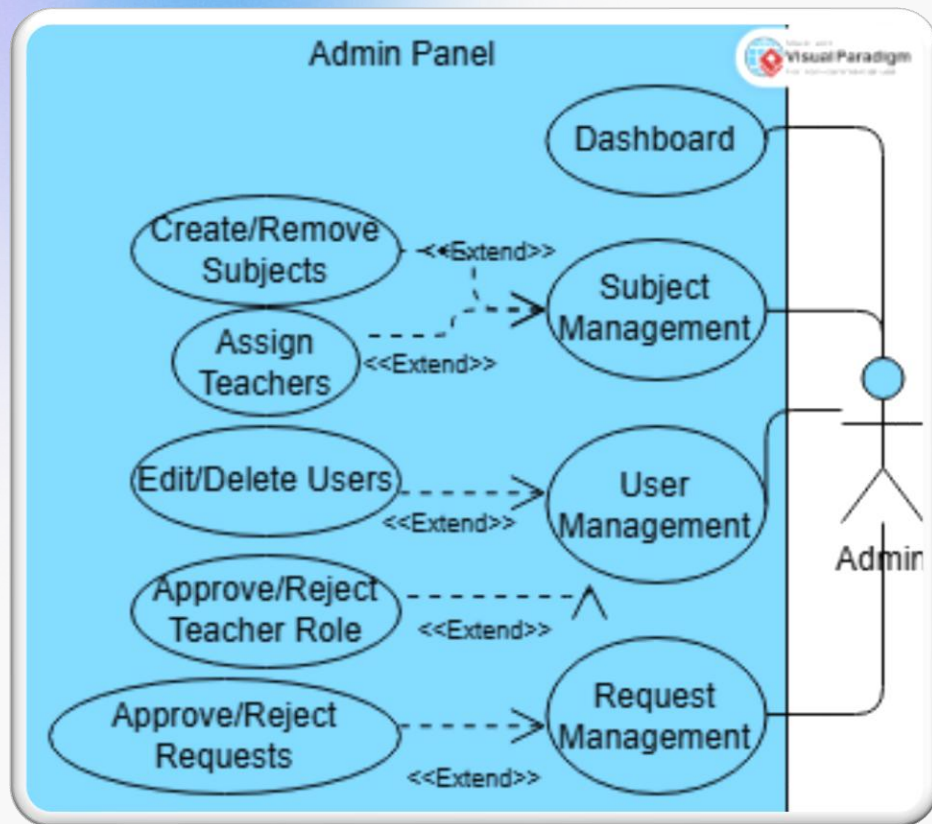
Professor

- Manages classes and student enrollments.
- Creates and organizes exercises and quizzes.
- Uploads and updates study materials (PDFs).
- Monitors class performance via the dashboard.

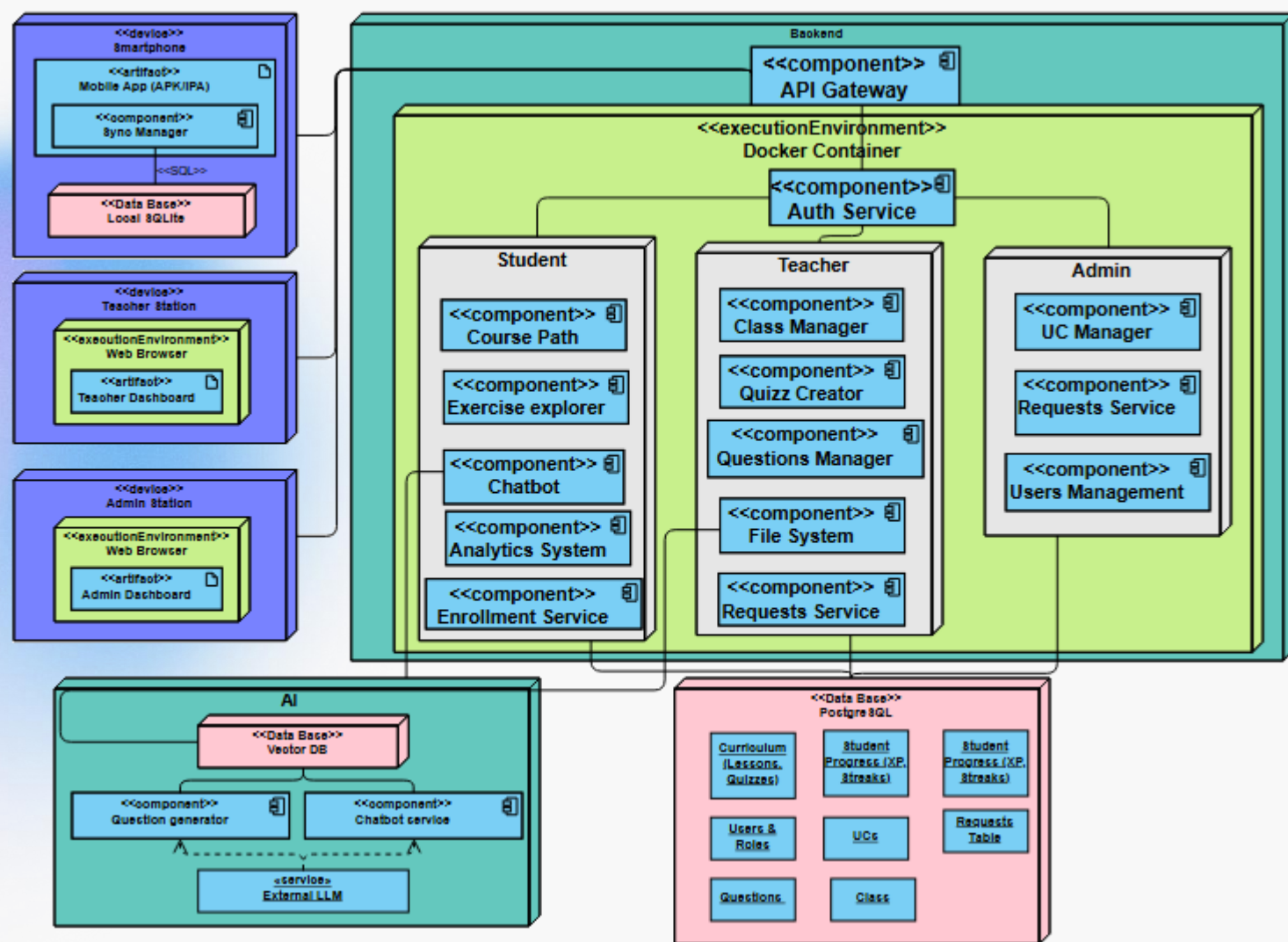


Admin

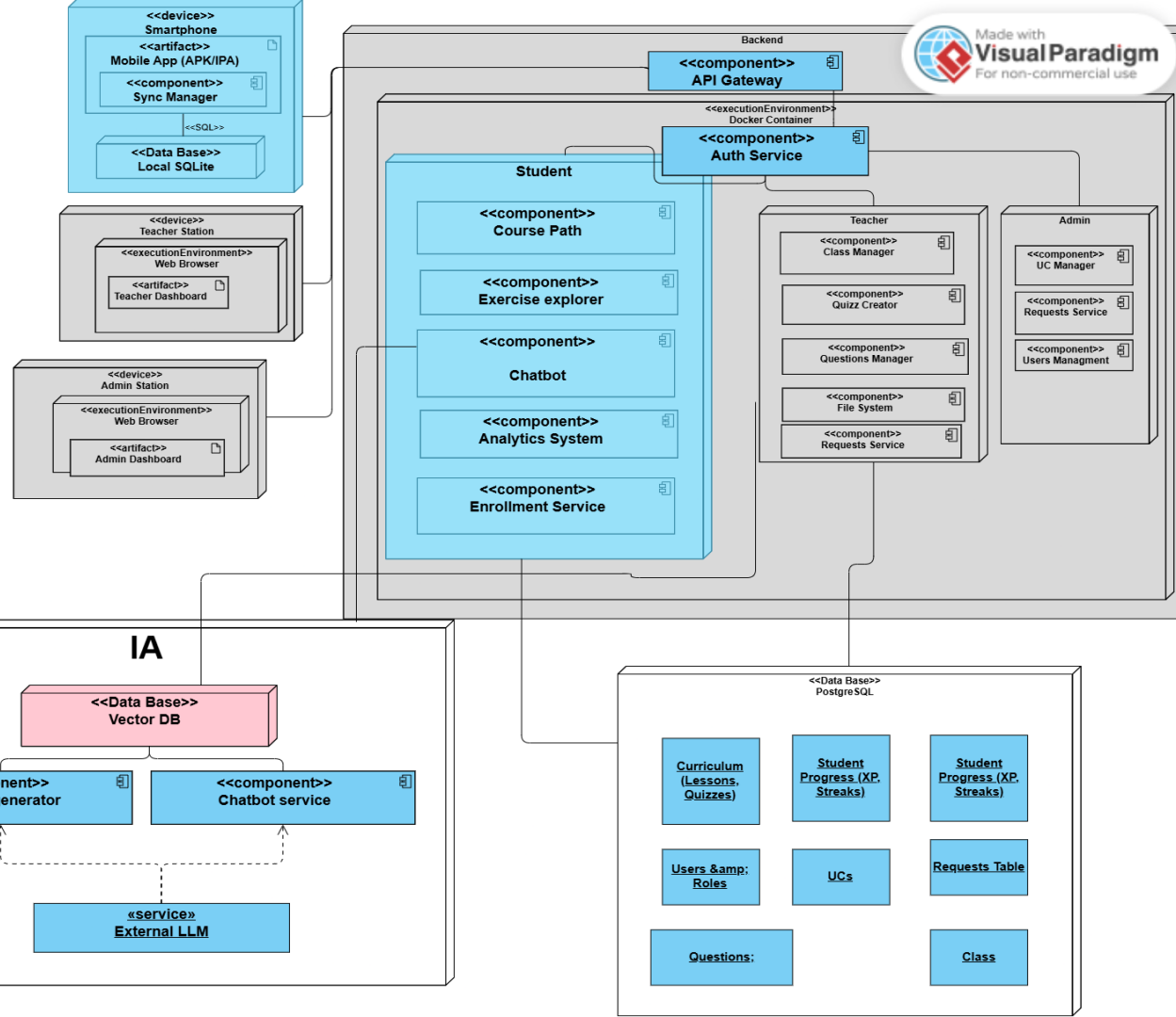
- Oversees the entire platform and user roles.
- Manages course/subject creation and teacher assignments.
- Handles system requests and permissions.
- Ensures the overall maintenance and security of the app.



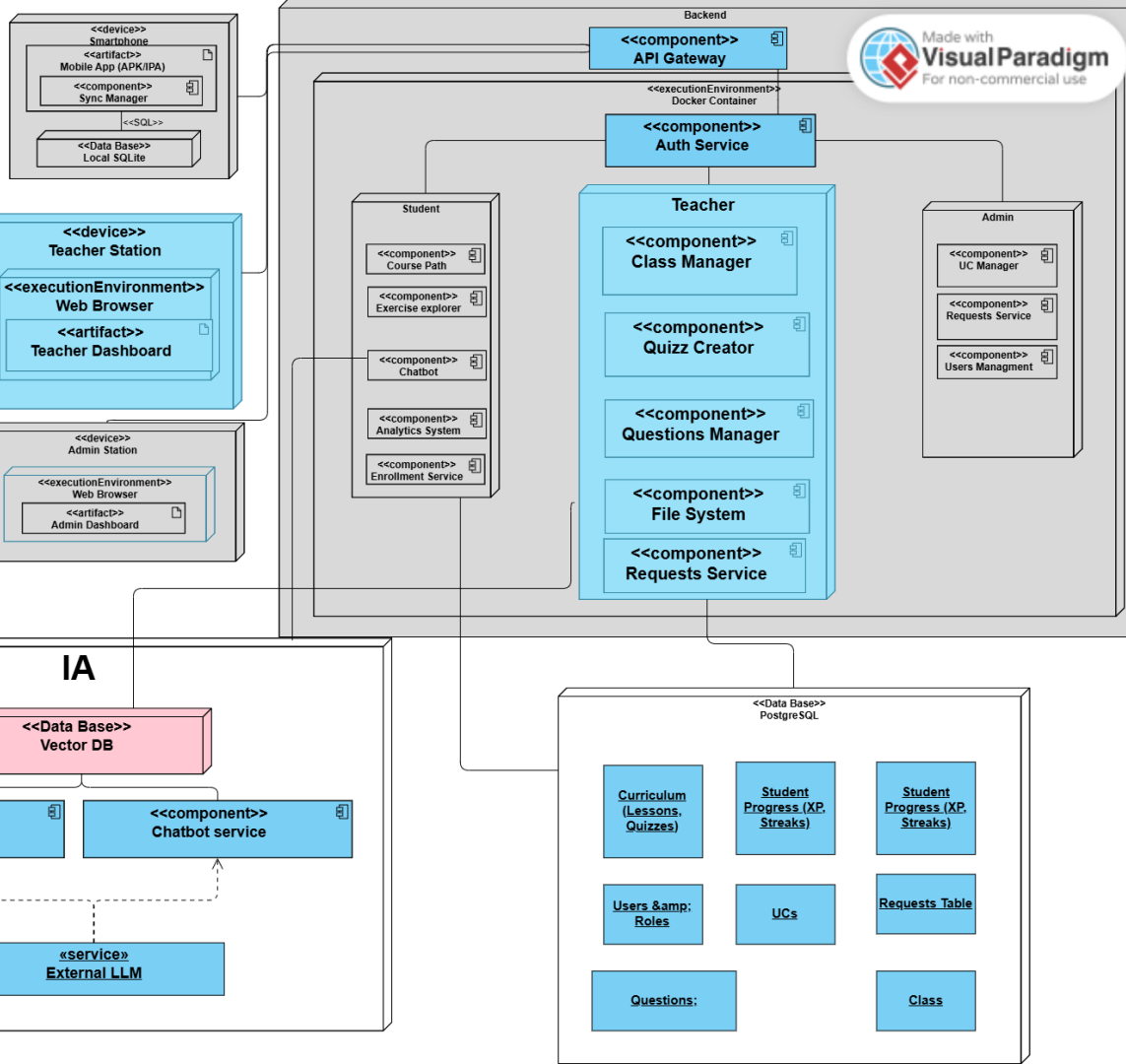
General Architecture



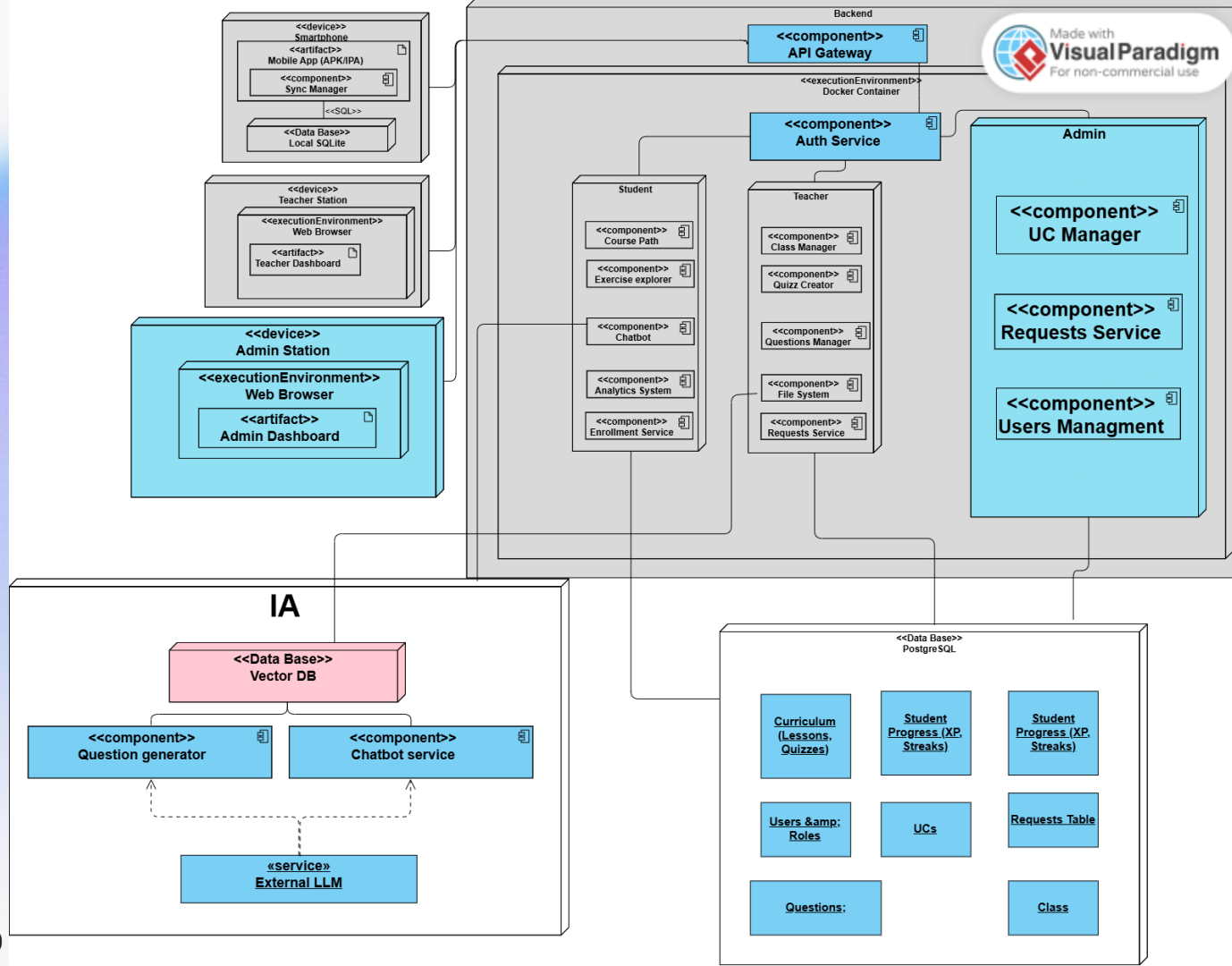
Student Focused Architecture



Teacher Focused Architecture



Admin Focused Architecture



Thank you

What's Next:

M3 - Mid Construction